

Homework 2

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MA726 Complex Geometry

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Problem 1 (2.3). Let $U \subseteq \mathbb{C}^n$ be an open set, and $f = f(z) \in \mathcal{O}(U)$ a non-zero holomorphic function. Denote by $X \subseteq U$ the complement of the zero-locus of f :

$$X = U \setminus Z(f) \subseteq U.$$

Show that X is isomorphic to a closed submanifold of $U \times \mathbb{C}$. (Hint: consider the function $f(z) \cdot t - 1 \in \mathcal{O}(U)[t]$.)

Proof. Let $p : U \times \mathbb{C} \rightarrow \mathbb{C}$ be $p(z, t) = f(z)t - 1$, which is clearly holomorphic on $U \times \mathbb{C} \subseteq \mathbb{C}^{n+1}$. We have $p(z, t) = 0$ whenever $f(z)t = 1$, hence the zero locus is

$$Z(p) = \{(z, t) \in U \times \mathbb{C} \mid f(z)t = 1\} \subseteq U \times \mathbb{C}.$$

This is closed since $\{0\} \subseteq \mathbb{C}$ is closed, since p is holomorphic (in particular, continuous), and since continuous preimages of closed sets are closed. We construct an isomorphism with X as follows. Let $\pi : U \times \mathbb{C} \rightarrow U$ be the canonical projection $(z, t) \mapsto z$, and write π_Z for the restriction $\pi_Z : Z(p) \rightarrow \pi(Z(p))$. Each $z \in \pi(Z(p))$ determines precisely one $t \in \mathbb{C}$ such that $(z, t) \in Z(p)$. Hence π_Z admits a continuous inverse

$$\pi_Z^{-1}(z) = (z, 1/f(z)).$$

Since $z \in \pi(Z(p))$, we have $f(z) \neq 0$, and thus $1/f(z)$ is always defined. Thus π_Z is a biholomorphism on $\pi(Z(p))$ since the inverse is holomorphic due to f being holomorphic.

To show $Z(p)$ is a complex manifold of dimension n , we appeal to Lazarsfeld example 2.6 (or more generally the implicit function theorem), which says we need only show there exists a coordinate z_i with $\frac{\partial p}{\partial z_i}(z) \neq 0$ for all $z \in Z(p)$. We have

$$\frac{\partial p}{\partial t} = f(z),$$

and as shown earlier f is nonzero everywhere on X . Thus $\partial_t p \neq 0$ on $Z(p)$, so $Z(p)$ is a complex submanifold of codimension 1. ■

Problem 2 (2.4). Let Y be a complex manifold of dimension $n + k$. A subset $X \subseteq Y$ is a closed n -submanifold if and only if one can find an open cover $\{U_\alpha\}_{\alpha \in A}$ of Y , together with a k -tuple $\{f_\alpha\}_{\alpha \in A}$ of holomorphic functions on each U_α , such that

$$U_\alpha \cap X = Z(f_\alpha),$$

and $J(f_\alpha)$ has maximal rank on $U_\alpha \cap X$.

Proof. ($\implies$) Let $X \subseteq Y$ be a closed submanifold. Then X is a closed $2n$ -submanifold of Y , and there is a holomorphic atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ on Y such that if $U_\alpha \cap X \neq \emptyset$, then

$$\varphi_\alpha : U_\alpha \cap X \cong \varphi_\alpha(U_\alpha) \cap \mathbb{C}^n$$

for some linear embedding $\mathbb{C}^n \cong \mathbb{C}^n \times \{0\} \hookrightarrow \mathbb{C}^{n+k}$. Write φ_α^i for the i th component function of φ_α such that

$$\varphi_\alpha(z) = (\varphi_\alpha^1(z), \dots, \varphi_\alpha^{n+k}(z)).$$

Since for $z \in X$ we have that $\varphi_\alpha(z) \in \varphi_\alpha(U_\alpha) \cap \mathbb{C}^n$, we have that only the first n components $\varphi_\alpha^{i \leq n}(z)$ are nonzero, and the final k components $\varphi_\alpha^{i > n}(z)$ must be zero. Consider the open cover $\{U_\alpha\}_{\alpha \in A}$ of Y together with the k -tuples of functions

$$\left\{ \left(\varphi_\alpha^{n+1}, \dots, \varphi_\alpha^{n+k} \right) \right\}_{\alpha \in A}.$$

Each function is immediately holomorphic since φ_α is holomorphic, and by definition of holomorphy.

For simplicity of notation we write $Z(\varphi_\alpha^i)$ to always mean the zero locus of the tuple $(\varphi_\alpha^{n+1}, \dots, \varphi_\alpha^{n+k})$. Let $z \in U_\alpha \cap X$. Then $\varphi_\alpha(z) \in \varphi_\alpha(U_\alpha) \cap \mathbb{C}^n$ implies that $\varphi_\alpha^i(z) = 0$ for all $n < i \leq n+k$, and thus $z \in Z(\varphi_\alpha^i)$. Conversely, let $z \in Z(\varphi_\alpha^i)$. Then $\varphi_\alpha(z) = (\varphi_\alpha^1(z), \dots, \varphi_\alpha^n(z), 0, \dots, 0)$, and thus $\varphi_\alpha(z) \in \mathbb{C}^n \times \{0\}$. It follows that $\varphi_\alpha(z) \in \varphi_\alpha(U_\alpha) \cap \mathbb{C}^n$, and since φ_α is an isomorphism, we obtain $z \in U_\alpha \cap X$. Thus we obtain $U_\alpha \cap X = Z(\varphi_\alpha^i)$.

It remains only to show that the Jacobian has maximal rank. We view the tuple of functions as a map $U_\alpha \rightarrow \mathbb{C}^k$ via $z \mapsto (\varphi_\alpha^i(z))_{n < i \leq n+k}$, which is holomorphic since each φ_α^i is holomorphic. The (real) Jacobian, which we simply write as $J(\varphi_\alpha^i)$ for simplicity of notation, is then of the form

$$J(\varphi_\alpha^i) = \begin{pmatrix} \frac{\partial \varphi_\alpha^{n+1}}{\partial z_{n+1}} & \dots & \frac{\partial \varphi_\alpha^{n+1}}{\partial z_{n+k}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \varphi_\alpha^{n+k}}{\partial z_{n+1}} & \dots & \frac{\partial \varphi_\alpha^{n+k}}{\partial z_{n+k}} \end{pmatrix},$$

where we note again that the complexified Jacobian is completely determined by the real Jacobian on holomorphic functions, and moreover attains maximal rank whenever the real one does. It is simple to show that the Jacobian contains the $k \times k$ identity matrix as a block matrix: since φ_α gives coordinates on U_α , the functions φ_α^i are precisely the coordinate functions, and thus $\partial_i \varphi_\alpha^j = \delta_{ij}$. Hence $J(\varphi_\alpha^i)$ contains the identity at all points on U_α , including on the intersection $U_\alpha \cap X$. It thus has rank at least k , which is the maximal rank since it maps from a $(n+k)$ -dimensional space to a k -dimensional space.

($\Leftarrow$) Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of Y , and $\{f_\alpha = (f_\alpha^1, \dots, f_\alpha^k)\}_{\alpha \in A}$ a collection of holomorphic functions on U_α such that $U_\alpha \cap X = Z(f_\alpha)$ and such that $J(f_\alpha)$ attains maximal rank on $U_\alpha \cap X$.

Since X is the zero locus of a collection of holomorphic functions, it is automatically closed. It is also a real submanifold of dimension $2n$ by the same theorem for smooth geometry. Let $\alpha \in A$ and let $p \in U_\alpha \cap X$. Choose a chart (V, z_i) with $p \in V \subseteq U_\alpha$, and for each f_α^i , view the Jacobian $J(f_\alpha^i)$ as an element of the tangent space¹ $T_p^{\mathbb{C}}V$ via the identification $\partial_{z_i} \leftrightarrow dz_i$. Since the Jacobian has maximal rank on $U_\alpha \cap X$, the set $\{J(f_\alpha^i)\}_{1 \leq i \leq k}$ is linearly independent in $T_p^{\mathbb{C}}V \cong T_p^{\mathbb{C}}U_\alpha$ for any $p \in U_\alpha \cap X$. Since derivatives with respect to coordinates span the tangent space, we can choose n coordinate functions $\{z_{i_j}\}_{1 \leq j \leq n}$ to complete the set of Jacobians to a basis $\left\{ J(f_\alpha^i), \frac{\partial}{\partial z_{i_j}} \right\}$ for the space.

Define the function $F = (z_{i_1}, \dots, z_{i_n}, f_\alpha^1, \dots, f_\alpha^k)$. Then since the components of F have linearly independent derivatives, the Jacobian of F is invertible. In particular, it is surjective for any $p \in U_\alpha \cap X$. By the inverse function theorem, there is an open neighborhood $p \in W_p \subseteq U_\alpha$ such

¹I did this because we haven't explicitly mentioned the cotangent space yet, but it would be more precise to say the df_α^i 's are linearly indendent in the holomorphic cotangent space.

that F restricts to an isomorphism $F : W_p \cong F(W_p)$. Since we can construct such a W_p for any $p \in X \cap U_\alpha$ and for any U_α , we immediately have that there is a holomorphic atlas containing $\{(W_p, F)\}_{p \in X}$. Moreover, by construction F restricted to X yields $F : W_p \cap X \cong F(W_p) \cap \mathbb{C}^n$ for any p , since the last k component functions are f_α^i , which vanish on X . It remains only to complete $\{(W_p, F)\}_{p \in X}$ into an atlas such that the only charts intersecting X are the W_p s. This is simple to do: choose a holomorphic atlas $\{(V_i, \varphi_i)\}_{i \in I}$ for Y , and since X is closed, its complement X^c is open. Then each $V_i \cap X^c$ is open, and the restriction $\varphi_i|_{V_i \cap X^c}$ is still a biholomorphism onto an open subset of $\mathbb{C}^{n+k}$. Thus we may construct the new atlas

$$\{(V_i \cap X^c, \varphi_i|_{V_i \cap X^c})\}_{i \in I} \cup \{(W_p, F)\}_{p \in X}.$$

As noted before, F is compatible with all coordinate functions, and thus this is indeed a holomorphic atlas. ■

Problem 3 (2.5). *Working in $\mathbb{C}^2$ with coordinates z, w , let*

$$X = \{w^2 - p_{2g+2}(z) = 0\}$$

be the curve defined as the zero locus of a polynomial $w^2 - p_{2g+2}(z)$, where p_{2g+2} is a polynomial of degree $2g + 2$ in z with distinct roots.

1. *Prove that the condition of Example 2.6 is met, and hence that X is a complex manifold of dimension 1.*
2. *By studying the projection*

$$\pi : X \rightarrow \mathbb{C}, \quad (z, w) \mapsto z$$

show that one can compactify X by adding two points “at infinity” to obtain a compact Riemann surface $\overline{X}$ in such a way that π extends to a holomorphic map

$$\overline{\pi} : \overline{X} \rightarrow S \cong \mathbb{C} \cup \{\infty\}.$$

(Hint: because p has even degree, the inverse image under π of the complement D^ of a disk of very large radius consists of the disjoint union of two copies of D^* . But D^* is a punctured neighborhood of the point at infinity, and therefore we need to add two points to compactify X .)*

It is instructive to think through why $\overline{X}$ is diffeomorphic to a compact surface of genus g . When $g \geq 2$, such a curve or Riemannian surface is called hyperelliptic.

Proof. (1) Example 2.6 says that given $U \subseteq \mathbb{C}^{n+1}$ and a holomorphic map $f : U \rightarrow \mathbb{C}$, if for all points $a \in Z(f)$ there is an index i such that $\frac{\partial f}{\partial z_i}(a) \neq 0$, then the zero locus $Z(f)$ is a complex manifold of dimension n . Since $p : \mathbb{C}^2 \rightarrow \mathbb{C}$ has the derivative $\frac{\partial p}{\partial w} = 2w$ which is only zero at $w = 0$, we need only check points $(w, z) = (0, z)$.

We show that if $p_{2g+2}(z_0) = 0$, then $p'_{2g+2}(z_0) \neq 0$. Suppose that z_0 is a root for p_{2g+2} . Then since p_{2g+2} splits over $\mathbb{C}$, it can be written

$$p_{2g+2}(z) = a(z - c_1) \cdots (z - c_{2g+2}) = a \prod_{i=1}^{2g+2} (z - c_i).$$

Since z_0 is a root, there exists k such that $c_k = z_0$. the product rule yields

$$\frac{\partial p_{2g+2}}{\partial z} = a \sum_{i=1}^{2g+2} \prod_{j \neq i} (z - c_j).$$

Evaluating at $z_0 = c_k$ yields

$$\frac{\partial p_{2g+2}}{\partial z}(c_k) = a \prod_{j \neq k} (c_k - c_j).$$

Here we note that every term other than the k th term of the sum is annihilated by the factor $(z - c_k)$. Since all roots are distinct, $c_k - c_j \neq 0$ for all $j \neq k$. Thus the product of these terms is nonzero, and thus $p'_{2g+2}(c_k) \neq 0$. Therefore we have that if $(0, z) \in X$, since this implies that $p_{2g+2}(z) = 0$, the partial with respect to z (which is $-p'_{2g+2}$) is therefore nonvanishing at $(0, z)$, and X is therefore a complex manifold of dimension 1.

(2) Recall (Lazardsfeld example 2.5) that the coordinate charts on the Riemann sphere $\mathbb{C} \cup \{\infty\}$ are given by $(\mathbb{C}, z)$ and $(\mathbb{C} \cup \{\infty\} \setminus \{0\}, u = 1/z)$ with $u(\infty) = 0$.

Let D be an *open* disk in $\mathbb{C}$ of radius R sufficiently large such that all roots of p_{2g+2} fall inside D . On the complement $D^* := \mathbb{C} \setminus D$, we can write

$$p_{2g+2}(z) = a_{2g+2}z^{2g+2} + \cdots + a_0 = a_{2g+2}z^{2g+2} \left(1 + \frac{a_{2g+1}}{a_{2g+2}z^{2g+2}} + \cdots + \frac{a_0}{a_{2g+2}z^{2g+2}} \right).$$

Write $H(z) = 1 + \cdots + \frac{a_0}{a_{2g+2}z^{2g+2}}$ such that $p_{2g+2}(z) = a_{2g+2}z^{2g+2}H(z)$ on D^* . We claim that we may choose R sufficiently large such that for some $0 < \varepsilon < 1$, the image of H falls within the ball $B_\varepsilon(1) \subseteq \mathbb{C}$. Importantly, the image of H does not contain 0 or complex numbers with negative real component. To show this, let

$$C := \max_{1 \leq i < 2g+2} \frac{a_i}{a_{2g+2}},$$

and thus for any $z \in D^*$,

$$\begin{aligned} |H(z) - 1| &= \left| \frac{a_{2g+1}}{a_{2g+2}z} + \cdots + \frac{a_0}{a_{2g+2}z^{2g+2}} \right| \\ &\leq \left| \frac{a_{2g+1}}{a_{2g+2}} \right| \frac{1}{|z|} + \cdots + \left| \frac{a_0}{a_{2g+2}} \right| \frac{1}{|z|^{2g+2}} \\ &\leq C \frac{1}{|z|} + \cdots + C \frac{1}{|z|^{2g+2}} \\ &\stackrel{!}{\leq} C \left(\frac{1}{|z|} + \cdots + \frac{1}{|z|} \right) \\ &= \frac{C(2g+2)}{|z|}, \end{aligned}$$

where the inequality marked by ! follows by taking $R > 1$. To take $H(z) \in B_\varepsilon(1)$, we must have $|H(z) - 1| < \varepsilon$, so we want to take $\frac{C(2g+2)}{|z|} < \varepsilon$. It follows that we must take $|z| > \frac{C(2g+2)}{\varepsilon}$, so simply take $R > \max \left\{ 1, \frac{C(2g+2)}{\varepsilon}, \text{magnitudes of zeroes of } p_{2g+2} \right\}$. This implies such an R exists for which the image of H falls in $B_\varepsilon(1)$ for any $0 < \varepsilon$; in particular for $\varepsilon < 1$.

For any $z \in D^*$, we can thus write $H(z) = |H(z)|e^{i\theta}$ for some $\theta \in (-\pi/2, \pi/2)$. Since $|H(z)| > 0$, we can then define a single-valued holomorphic principal square root function on its image:

$$\sqrt{H(z)} = \sqrt{|H(z)|}e^{i\theta/2}.$$

by taking a single branch of the real square root function $\sqrt{|H(z)|}$. Since the degree of p_{2g+2} is even, we can extend this to define

$$f_+(z) = z^{g+1} \sqrt{a_{2g+2}} \sqrt{H(z)},$$

$$f_-(z) = -z^{g+1} \sqrt{a_{2g+2}} \sqrt{H(z)},$$

where we choose some value of $\sqrt{a_{2g+2}}$.

Now note that any fiber of $z \in D^*$ under the projection can be described as

$$\begin{aligned} \pi^{-1}(z) &= \{(z, w) \in \mathbb{C}^2 \mid w^2 = p_{2g+2}(z)\} = \left\{ \left(z, \sqrt{p_{2g+2}(z)} \right), \left(z, -\sqrt{p_{2g+2}(z)} \right) \right\} \\ &= \{(z, f_+(z)), (z, f_-(z))\}. \end{aligned}$$

Since these are distinct when $z \in D^*$, we obtain that the preimage $\pi^{-1}(D^*)$ can be described as the disjoint union of the two components

$$\begin{aligned} U_+ &:= \{(z, f_+(z)) \mid z \in D^*\}, \\ U_- &:= \{(z, f_-(z)) \mid z \in D^*\}. \end{aligned}$$

Note also that since $f_{\pm}$ are holomorphic, and the map $z \mapsto (z, f_{\pm}(z))$ admits a holomorphic inverse by restricting the projection $\pi|_{U_{\pm}}$, we have isomorphisms $U_+ \cong D^* \cong U_-$. To compactify D^* , we add a point at ∞ , and coordinates near ∞ are defined as $u = 1/z$ by taking $D^* \cup \{\infty\}$ to be a submanifold of $\mathbb{C} \cup \{\infty\}$. Hence to compactify $U_{\pm}$, we must add two points $U_+^{\infty} := U_+ \cup \{\infty_+\}$ and $U_-^{\infty} := U_- \cup \{\infty_-\}$. We define $\bar{X} := X \cup \{\infty_+, \infty_-\}$, and define charts at these points by defining charts on some open neighborhood² $V_{\pm} \subseteq U_{\pm}^{\infty}$ of the points $\infty_{\pm}$ by mapping $U_{\pm}$ along the isomorphism to D^* , mapping $\infty_{\pm} \mapsto \infty$, and then taking the chart there. Since these maps are isomorphisms, they are automatically compatible with the chart on X , hence we have a holomorphic atlas³ on $\bar{X}$.

Define $\bar{\pi} : \bar{X} \rightarrow \mathbb{C} \cup \{\infty\}$ by $\infty_{\pm} \mapsto \infty$ and $\bar{\pi}|_X = \pi$. We only need to show $\bar{\pi}$ is holomorphic near $\infty_{\pm}$, so without loss of generality let $p \in V_+$. Write φ_+ for the coordinate chart on V_+ , which we have defined to be equal to $p \mapsto 1/\pi(p)$, where we note once again that $1/\infty$ is defined to be 0. Write $\psi(z) = 1/z$ for the coordinates on $\mathbb{C} \cup \{\infty\}$ near ∞ , such that the coordinate representation $\psi \circ \bar{\pi} \circ \varphi_+^{-1}$ is given by

$$\frac{1}{\bar{\pi}(p)} \xrightarrow{\varphi_+^{-1}} p \xrightarrow{\bar{\pi}} \bar{\pi}(p) \xrightarrow{\psi} \frac{1}{\bar{\pi}(p)}.$$

This is the identity map, which is holomorphic.

Finally, although each $U_{\pm}^{\infty} \subseteq \bar{X}$ is compact, we have not yet shown that $\bar{X}$ is compact. This is fairly simple to do: since a finite union of compact sets is compact, we simply need to find a compact subset Y of X such that $U_+^{\infty} \cup U_-^{\infty} \cup Y = \bar{X}$. Fix some finite real number $R_0 > R$, and define $Y := \pi^{-1}(\overline{B_{R_0}(0)})$. Polynomials are continuous and bounded on compact sets, and thus so is their square root. Therefore Y is bounded, and since it is a continuous preimage of a closed set, it is closed. By Heine-Borel, Y is compact. To show $\bar{X} = U_+^{\infty} \cup U_-^{\infty} \cup Y$, suppose that there exists a point p not in Y . We can take $p \neq \infty_{\pm}$ since those fall in $U_{\pm}^{\infty}$. Since $p \notin Y = \pi^{-1}(\overline{B_{R_0}(0)})$, we note that $|\pi(p)| \geq R_0 > R$. But then $\pi(p) \in D^*$, and the preimage of this is $U_+ \cup U_-$, which is a subset of $U_+^{\infty} \cup U_-^{\infty}$. Therefore p is in the union, so $\bar{X}$ is the finite union of compact spaces, and thus compact. $\blacksquare$

²Each $U_{\pm}^{\infty}$ is closed, so this is necessary. This is an artifact of choosing D to be open, which simplifies the compactness proof.

³I was concerned that assuming this atlas was holomorphically compatible at the points at ∞ was assuming that $\bar{\pi}$ was holomorphic there, but the holomorphic atlas only needs to be compatible on the intersections of charts, which do not include the points at ∞ . Thus one only needs $\bar{\pi}$ to be continuous at $\infty_{\pm}$ for this step, which is immediate.

Problem 4 (2.8). *Let*

$$\Lambda_0 = \mathbb{Z}^n + i\mathbb{Z}^n \subseteq \mathbb{C}^n$$

and let $\Lambda \subseteq \mathbb{C}^n$ be a lattice spanned by $2n$ vectors $\{\lambda_j\}_{1 \leq j \leq 2n} \subseteq \mathbb{C}^n$. Assume that the $2n^2$ complex numbers appearing as coordinates of the λ_j 's are algebraically independent over $\mathbb{Q}$. Then Λ_0 and Λ determine non-isomorphic complex tori.

Proof. A morphism $f : X_{\Lambda_0} \rightarrow X_{\Lambda}$ is fully and uniquely determined by a $\mathbb{C}$ -linear map $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ mapping $A(\Lambda_0) \subseteq \Lambda$. It's fairly immediate that if f is an isomorphism, then A must be an isomorphism, and thus $A(\Lambda_0) = \Lambda$. Thus it suffices to show there is no $\mathbb{C}$ -linear isomorphism $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ mapping $A(\Lambda_0) = \Lambda$.

By hypothesis, the $2n^2$ components of the λ_j 's are not the roots of a single nontrivial polynomial with coefficients in $\mathbb{Q}$. Suppose for contradiction that such an A exists; we will show that the existence of A determines a polynomial p in $2n^2$ variables with coefficients in $\mathbb{Q}$ for which the set S of components of the λ_j 's is a root.

Write e_i for the i th basis vector of $\mathbb{R}^n$ (equivalently $\mathbb{C}^n$), and note that Λ_0 is a free $\mathbb{Z}$ -module generated by $\{e_j, ie_j\}_{1 \leq j \leq n}$. Since $A(\Lambda_0) = \Lambda$ and A is $\mathbb{C}$ -linear, it follows that Λ is also spanned by $\{A(e_j), iA(e_j)\}_{1 \leq j \leq n}$ as a $\mathbb{Z}$ -module. It follows by injectivity of A that this set is a $\mathbb{Z}$ -basis, hence A determines an isomorphism $\Lambda_0 \cong \Lambda$ as $\mathbb{Z}$ -modules with $\{A(e_j), iA(e_j)\}_{1 \leq j \leq n}$ a basis for Λ . Thus for each $0 \leq j \leq 2n$, there exist integers c_j^k , $0 \leq k \leq 2n$ such that

$$\begin{aligned} \lambda_j &= c_j^1 A(e_1) + \cdots + c_j^n A(e_n) + c_j^{n+1} iA(e_1) + \cdots + c_j^{2n} iA(e_n) \\ &= \sum_{1 \leq k \leq n} \left(c_j^k + ic_j^{n+k} \right) A(e_k). \end{aligned}$$

For simplicity of notation, write $Z_{jk} = c_j^k + ic_j^{n+k} \in \mathbb{Z}[i]$, such that

$$\lambda_j = \sum_{1 \leq k \leq n} Z_{jk} A(e_k).$$

We can write A as a matrix in $\text{GL}_n(\mathbb{C})$ with respect to the $\{e_j\}_{1 \leq j \leq n}$ basis for $\mathbb{C}$. Write A_{jk} for the jk th entry of A . Let V be the $\mathbb{Q}(i)$ -vector space spanned by the matrix entries $\{A_{jk}\}_{1 \leq j, k \leq n}$ of A . Since there are n^2 such entries, note that $\dim_{\mathbb{Q}(i)} V \leq n^2$. Write λ_j^m for the m th entry of λ_j . Then since

$$\lambda_j = \sum_{1 \leq k \leq n} Z_{jk} A(e_k) = \sum_{1 \leq k \leq n} \sum_{1 \leq m \leq n} Z_{jk} A_{mk} e_k,$$

we must have that

$$\lambda_j^m = \sum_{1 \leq k \leq n} Z_{jk} A_{mk}.$$

Since each Z_{jk} is an element of $\mathbb{Z}[i]$, it is in particular an element of $\mathbb{Q}(i)$. Thus all $2n^2$ elements of $S := \left\{ \lambda_j^m \right\}_{j,m}$ belong to the $\mathbb{Q}(i)$ -vector space V . Since S has $2n^2$ elements, which is more than the dimension of the space, S must be linearly dependent in V . Thus there are coefficients $a_{jm} \in \mathbb{Q}(i)$ for all $1 \leq m \leq n$ and $1 \leq j \leq 2n$ which are *not all zero*, such that

$$\sum_{j=1}^{2n} \sum_{m=1}^n a_{jm} \lambda_j^m = 0.$$

Write $a_{jm} = u_{jm} + iv_{jm}$ with $u_{jm}, v_{jm} \in \mathbb{Q}$ not all zero, such that

$$\sum_{j,m} (u_{jm} + iv_{jm}) \lambda_j^m = 0.$$

To turn this into a polynomial in $\mathbb{Q}$, we multiply by the complex conjugate

$$\left(\sum_{j,m} (u_{jm} + iv_{jm}) \lambda_j^m \right) \left(\sum_{j,m} (u_{jm} - iv_{jm}) \lambda_j^m \right) = 0.$$

By linearity of the complex conjugate, and by $z\bar{z} = |z|^2$, we obtain that this equation is

$$\left(\sum_{j,m} u_{jm} \lambda_j^m \right)^2 + \left(\sum_{j,m} v_{jm} \lambda_j^m \right)^2 = 0.$$

Define the polynomial $p \in \mathbb{Q}[\{x_{jm}\}_{j,m}]$ in $2n^2$ variables over $\mathbb{Q}$ as

$$p(\{x_{jm}\}_{j,m}) = \left(\sum_{j,m} u_{jm} x_{jm} \right)^2 + \left(\sum_{j,m} v_{jm} x_{jm} \right)^2 = 0.$$

Since $\{\lambda_j^m\}_{j,m}$ is a solution to this polynomial, and since not all coefficients of the polynomial are zero, there exists a nontrivial polynomial p in $\mathbb{Q}$ for which the entries of the λ_j 's are a root, contradicting their algebraic independence over $\mathbb{Q}$. Therefore no isomorphism of complex tori exists. ■